

MATHEMATICS STANDARD 2026 ALL SETS

CHAPTER 1 - REAL NUMBERS

=====

CHAPTER: 1 - Real Numbers

=====

YEAR: 2025-26
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The LCM of 960 and 240 is:
OPTIONS: (A) 960 (B) 240 (C) 60 (D) 15
ANSWER: (A) 960

YEAR: 2025-26
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The natural number 1 is:
OPTIONS: (A) a prime number. (B) a composite number. (C) prime as well as composite. (D) neither prime nor composite.
ANSWER: (D) neither prime nor composite.

YEAR: 2025-26
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: For any natural number n , 5^n ends with the digit:
OPTIONS: (A) 0 (B) 5 (C) 3 (D) 2
ANSWER: (B) 5

YEAR: 2025-26
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1

QUESTION: For any natural number n , 6^n ends with the digit:

OPTIONS: (A) 0 (B) 6 (C) 3 (D) 2

ANSWER: (B) 6

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: $(3 \times 11 \times 13 + 3)$ is:

OPTIONS: (A) a prime number (B) divisible by 13 (C) a composite number (D) an odd number

ANSWER: (C) a composite number

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: The HCF of 960 and 432 is:

OPTIONS: (A) 48 (B) 54 (C) 72 (D) 36

ANSWER: (A) 48

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The natural number 2 is:

OPTIONS: (A) a prime number (B) a composite number (C) prime as well as composite (D) neither prime nor composite

ANSWER: (A) a prime number

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The HCF of 960 and 432 is:

OPTIONS: (A) 48 (B) 54 (C) 72 (D) 36

ANSWER: (A) 48

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: There are two sections A and B of Grade X. There are 28 students in Section A and 30 students in Section B. What is the minimum number of books you will acquire for the class library so that they can be distributed equally among students of Section A or Section B?

OPTIONS: (A) 144 (B) 2 (C) 420 (D) 272

ANSWER: (C) 420

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the length of the plank that can be used to measure the lengths 4 m 20 cm and 5 m 4 cm exactly, in the least time.

OPTIONS: (A) 84 cm (B) 72 cm (C) 96 cm (D) 108 cm

ANSWER: (A) 84 cm

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If the HCF of 210 and 55 is expressed as $210 \times 5 + 55m$, then find the value of m .

OPTIONS: (A) -19 (B) -17 (C) 19 (D) 17

ANSWER: (A) -19

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the greatest number less than 10,000 which is exactly divisible by 48, 60 and 65.

OPTIONS: (A) 9360 (B) 6240 (C) 3120 (D) 9360

ANSWER: (A) 9360

YEAR: 2025-26

CHAPTER: 1 - Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A trader has three different types of oils of volume 870 l, 812 l and 638 l. Find the least number of containers of equal size required to store all the oil without getting mixed.

OPTIONS: (A) 40 (B) 35 (C) 42 (D) 38

ANSWER: (A) 40

CHAPTER 2 - POLYNOMIALS

=====

CHAPTER: 2 - Polynomials

=====

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The graph of $y = f(x)$ is given. The number of distinct zeroes of $y = f(x)$ is:

OPTIONS: (A) 0 (B) 1 (C) 2 (D) 3

ANSWER: (C) 2

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If α and β are two zeroes of a polynomial $f(x) = px^2 - 2x + 3p$ and $\alpha + \beta = \alpha\beta$, then value of p is:

OPTIONS: (A) $-\frac{2}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

ANSWER: (B) $\frac{2}{3}$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Observe the graph of polynomial $p(x)$. Number of zeroes of $p(x)$ is:

OPTIONS: (A) 5 (B) 4 (C) 6 (D) 3

ANSWER: (D) 3

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: How many zeroes does $p(x) = (x - 2)(x + 3)$ have?

OPTIONS: (A) Zero (B) One (C) Two (D) Three

ANSWER: (C) Two

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The sum and product of zeroes of a quadratic polynomial $p(x)$ are $-1/3$ and 2 respectively. The polynomial $p(x)$ is:

OPTIONS: (A) $3x^2 - x + 6$ (B) $x^2 + 1/3 x - 2$ (C) $3x^2 - x + 2$ (D) $-3x^2 - x - 6$

ANSWER: (D) $-3x^2 - x - 6$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A polynomial $p(x)$, which has sum of its zeroes equal to their product, is:

OPTIONS: (A) $3x^2 + 2x + 2$ (B) $3x^2 - 2x - 3$ (C) $3x^2 - 2x + 2$ (D) $x^2 - 3x + 2$

ANSWER: (C) $3x^2 - 2x + 2$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Observe the graph of polynomial $p(x)$. The zeroes of the polynomial are:

OPTIONS: (A) -2, 0, 2.5 (B) -2, 2.5 (C) 0, 4 (D) -2, 0

ANSWER: (B) -2, 2.5

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If α , β are the zeroes of the quadratic polynomial $px^2 + qx + r$, then find the value of $\alpha^3\beta + \beta^3\alpha$.

OPTIONS: (A) $r/p (q^2 - 2pr)$ (B) $r/p (q^2 + 2pr)$ (C) $q/p (r^2 - 2pq)$ (D) $r/p (q^2 - 2pr)$

ANSWER: (A) $r/p (q^2 - 2pr)$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If α , β are the zeroes of the polynomial $p(x) = x^2 - 3x - 1$, then find the value of $1/\alpha + 1/\beta$.

OPTIONS: (A) -3 (B) 3 (C) 1 (D) -1

ANSWER: (A) -3

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the value of p , for which one zero of the quadratic polynomial $px^2 - 14x + 8$ is 6 times the other.

OPTIONS: (A) 3 (B) 6 (C) 12 (D) 24

ANSWER: (A) 3

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the zeroes of the quadratic polynomial $x^2 + 7x + 10$, and verify the relationship between the zeroes and its coefficients.

OPTIONS: (A) -2 and -5 (B) 2 and 5 (C) -2 and 5 (D) 2 and -5

ANSWER: (A) -2 and -5

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the zeroes of a polynomial $p(x)$ are -3 and 8, then $p(x)$ equals:

OPTIONS: (A) $x^2 + 5x - 4$ (B) $(x + 3)(-x + 8)$ (C) $a(x^2 + 5x - 24)$ (D) $x^2 - 24$

ANSWER: (B) $(x + 3)(-x + 8)$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If sum and product of zeroes of a polynomial are (-3) and (-2) respectively, then a polynomial is:

OPTIONS: (A) $x^2 - 3x - 2$ (B) $-x^2 - 3x + 2$ (C) $-x^2 + 3x - 2$ (D) $x^2 + 3x + 2$

ANSWER: (B) $-x^2 - 3x + 2$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: α and β are the zeroes of the polynomial $5x^2 - 16x - 10$. Find the value of $\alpha/\beta + \beta/\alpha$.

OPTIONS: (A) $-356/50$ (B) $356/50$ (C) $-178/25$ (D) $178/25$

ANSWER: (C) $-178/25$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find a quadratic polynomial whose zeroes are $(5 - 2\sqrt{3})$ and $(5 + 2\sqrt{3})$.

OPTIONS: (A) $x^2 - 10x + 13$ (B) $x^2 + 10x + 13$ (C) $x^2 - 10x - 13$ (D) $x^2 + 10x - 13$

ANSWER: (A) $x^2 - 10x + 13$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: α and β are zeroes of the polynomial $p(x) = 3x^2 - 6x - 5$. Find the value of $1/\alpha^2 + 1/\beta^2$.

OPTIONS: (A) $66/25$ (B) $66/-25$ (C) $25/66$ (D) $-25/66$

ANSWER: (A) $66/25$

YEAR: 2025-26

CHAPTER: 2 - Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): The polynomial $p(y) = y^2 + 4y + 3$ has two zeroes. Reason (R): A quadratic polynomial can have at most two zeroes.

OPTIONS: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.

ANSWER: (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

CHAPTER 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

=====

CHAPTER: 3 - Pair of Linear Equations in Two Variables

=====

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the pair of linear equations: $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is consistent and dependent, then:

OPTIONS: (A) $a_1/a_2 \neq b_1/b_2$ (B) $a_1/a_2 \neq b_1/b_2 = c_1/c_2$ (C) $a_1/a_2 = b_1/b_2 \neq c_1/c_2$ (D) $a_1/a_2 = b_1/b_2 = c_1/c_2$

ANSWER: (D) $a_1/a_2 = b_1/b_2 = c_1/c_2$

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The pair of linear equations $3x/2 + 5y/3 = 7$ and $9x + 10y = 14$, is:

OPTIONS: (A) consistent (B) inconsistent (C) consistent with one solution (D) consistent with many solutions

ANSWER: (B) Inconsistent

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Equation of another line parallel to the line represented by $2x - 6y = 7$ is:

OPTIONS: (A) $y = 3x - 7$ (B) $2x = 9 - 6y$ (C) $x - 3y = 7$ (D) $x = 7/2 - 3y$

ANSWER: (C) $x - 3y = 7$

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If a pair of linear equations in two variables is represented by two coincident lines, then the pair of equations has:

OPTIONS: (A) a unique solution (B) two solutions (C) no solution (D) an infinite number of solutions

ANSWER: (D) an infinite number of solutions

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of k for which the system of linear equations $x/2 + y/3 = 5$ and $2x + ky = 7$ is inconsistent, is:

OPTIONS: (A) $3/4$ (B) $4/3$ (C) $1/3$ (D) 3

ANSWER: (B) $4/3$

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of k for which the system of linear equations $kx - y - 2 = 0$ and $6x - 2y - 3 = 0$ has infinitely many solutions, is (does):

OPTIONS: (A) $\frac{1}{2}$ (B) 3 (C) 4 (D) Not exist

ANSWER: (D) Not exist

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Aarush bought 2 pencils and 3 chocolates for Rs 11 and Tanish bought 1 pencil and 2 chocolates for Rs 7 from the same shop. Represent this situation in the form of a pair of linear equations. Find the price of 1 pencil and 1 chocolate, graphically.

OPTIONS: (A) Pencil = Rs 1, Chocolate = Rs 3 (B) Pencil = Rs 2, Chocolate = Rs 3 (C) Pencil = Rs 1, Chocolate = Rs 2 (D) Pencil = Rs 3, Chocolate = Rs 1

ANSWER: (A) Pencil = Rs 1, Chocolate = Rs 3

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Determine graphically, the coordinates of vertices of a triangle whose equations are $2x - 3y + 6 = 0$; $2x + 3y - 18 = 0$ and $x = 0$. Also, find the area of this triangle.

OPTIONS: (A) Vertices: (0,2), (3,4), (0,6); Area = 6 sq. units (B) Vertices: (0,2), (4,3), (0,6); Area = 8 sq. units (C) Vertices: (0,2), (3,4), (0,6); Area = 8 sq. units (D) Vertices: (2,0), (4,3), (6,0); Area = 6 sq. units

ANSWER: (A) Vertices: (0,2), (3,4), (0,6); Area = 6 sq. units

YEAR: 2025-26

CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A faster train takes one hour less than a slower train for a journey of 200 km. If the speed of the slower train is 10 km/hr less than that of the faster train, find the speeds of the two trains.

OPTIONS: (A) 50 km/h and 40 km/h (B) 60 km/h and 50 km/h (C) 45 km/h and 35 km/h (D) 55 km/h and 45 km/h

ANSWER: (A) 50 km/h and 40 km/h

CHAPTER 4 - QUADRATIC EQUATIONS

=====

CHAPTER: 4 - Quadratic Equations

=====

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The roots of the quadratic equation $(x - 1)^2 = 16$ are:

OPTIONS: (A) 5, 3 (B) 4, -4 (C) 5, -3 (D) -5, 3

ANSWER: (C) 5, -3

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the roots of the quadratic equation $\sqrt{3}x^2 - kx + 2\sqrt{3} = 0$ are real and equal, then the value(s) of k is/are:

OPTIONS: (A) $\pm\sqrt{24}$ (B) 0 (C) 4 (D) -5

ANSWER: (A) $\pm\sqrt{24}$

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the quadratic equation $9x^2 + 8kx + 16 = 0$ has real and equal roots, then the value of k is:

OPTIONS: (A) 3 (B) -3 (C) -4 (D) $\frac{3}{2}$

ANSWER: (A) 3 OR (B) -3

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If roots of the quadratic equation $x^2 - k\sqrt{3}x + 2 = 0$ are real and equal, then value of k is:

OPTIONS: (A) -2 (B) $\sqrt{8/3}$ (C) 1 (D) 2

ANSWER: (B) $\sqrt{8/3}$

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of k for which the equation $kx^2 - 6x - 4 = 0$ has real and equal roots, is:

OPTIONS: (A) $9/4$ (B) -4 (C) $-9/4$ (D) -2

ANSWER: (C) $-9/4$

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of m for which the quadratic equation $3x^2 - 7x + m = 0$ has real and equal roots, is:

OPTIONS: (A) 7 (B) $49/12$ (C) $49/3$ (D) 4

ANSWER: (B) $49/12$

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Verify that roots of the quadratic equation $(p - q)x^2 + (q - r)x + (r - p) = 0$ are equal when $q + r = 2p$.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A person on tour has Rs 5,400 for his expenses. If he extends his tour by 5 days, he has to cut down his daily expenses by Rs 180. Find the original duration of the tour and daily expense.

OPTIONS: (A) 10 days, Rs 540 (B) 12 days, Rs 450 (C) 15 days, Rs 360 (D) 8 days, Rs 675

ANSWER: (A) 10 days, Rs 540

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The total cost of certain piece of cloth was Rs 2,100. During special sale time, the shopkeeper offered 2 m extra cloth for free thus reducing the price of cloth per metre by Rs 120. What was the original per metre price of cloth and its length?

OPTIONS: (A) Rs 420, 5 m (B) Rs 350, 6 m (C) Rs 525, 4 m (D) Rs 300, 7 m

ANSWER: (A) Rs 420, 5 m

YEAR: 2025-26

CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Two pipes are used to fill a swimming pool. If the pipe of the larger diameter is used for 4 hours and the pipe of the smaller diameter for 9 hours, only half of the pool can be filled. Find how long it would take for each pipe to fill the pool, separately, if the pipe of smaller diameter takes 10 hours more than the pipe of larger diameter to fill the pool.

OPTIONS: (A) 20 hours and 30 hours (B) 15 hours and 25 hours (C) 25 hours and 35 hours (D) 18 hours and 28 hours

ANSWER: (A) 20 hours and 30 hours

CHAPTER 5 - ARITHMETIC PROGRESSIONS

=====

CHAPTER: 5 - Arithmetic Progressions

=====

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following sequence is not an A.P.?

OPTIONS: (A) 2, $\frac{5}{2}$, 3, $\frac{7}{2}$, ... (B) -1.2, -3.2, -5.2, -7.2, ... (C) $\sqrt{2}$, $\sqrt{8}$, $\sqrt{18}$, ... (D) 1^2 , 3^2 , 5^2 , 7^2 , ...

ANSWER: (D) 1^2 , 3^2 , 5^2 , 7^2 , ...

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The nth term of the A.P. $-\frac{1}{3}$, $\frac{2}{3}$, $\frac{5}{3}$, $\frac{8}{3}$, ... is:

OPTIONS: (A) $3n - 4$ (B) $n - \frac{4}{3}$ (C) $(n - 2)/3$ (D) $(n - 4)/3$

ANSWER: (B) $n - \frac{4}{3}$

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The common difference of the AP: $\sqrt{2}$, $2\sqrt{2}$, $3\sqrt{2}$, $4\sqrt{2}$, ... is:

OPTIONS: (A) $\sqrt{2}$ (B) 1 (C) $2\sqrt{2}$ (D) $-\sqrt{2}$

ANSWER: (A) $\sqrt{2}$

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The first term of an AP is p and the common difference is q, then its 10th term is:

OPTIONS: (A) $q - 9p$ (B) $p - 9q$ (C) $p + 9q$ (D) $2p + 9q$

ANSWER: (C) $p + 9q$

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If sum of first ten terms of an A.P. is zero with a as the first term and d , the common difference, which of the following relation is true?

OPTIONS: (A) $10a + 9d = 0$ (B) $2a = 9d$ (C) $a_{10} = -a$ (D) $a_{10} = a$

ANSWER: (C) $a_{10} = -a$

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In an A.P., the first term is 32 and the last term is -10. If the common difference is -2, then find the number of terms and their sum.

OPTIONS: (A) 22 terms, sum = 242 (B) 20 terms, sum = 220 (C) 24 terms, sum = 264 (D) 18 terms, sum = 198

ANSWER: (A) 22 terms, sum = 242

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the sum of the first 28 terms of an A.P. whose n th term is given by $a_n = 3n - 2$.

OPTIONS: (A) 1162 (B) 1120 (C) 1184 (D) 1150

ANSWER: (A) 1162

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In an A.P., the first term is 4 and the last term is 31. If sum of all the terms is 175, find the number of terms and the common difference.

OPTIONS: (A) 10 terms, $d = 3$ (B) 12 terms, $d = 2$ (C) 8 terms, $d = 4$ (D) 10 terms, $d = 2$

ANSWER: (A) 10 terms, $d = 3$

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: How many terms of the A.P. 21, 18, 15, ... must be added to get the sum zero?

OPTIONS: (A) 15 (B) 14 (C) 16 (D) 13

ANSWER: (A) 15

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In an A.P., 15th term exceeds the 8th term by 21. If sum of first 10 terms is 55, then form the A.P.

OPTIONS: (A) -8, -5, -2, ... (B) -7, -4, -1, ... (C) -9, -6, -3, ... (D) -8, -4, 0, ...

ANSWER: (A) -8, -5, -2, ...

YEAR: 2025-26

CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The sum of first n terms of an A.P. is $2n^2 + 13n$. Find its n th term and hence 10th term.

OPTIONS: (A) $a_n = 4n + 11$, $a_{10} = 51$ (B) $a_n = 4n + 13$, $a_{10} = 53$ (C) $a_n = 4n + 9$, $a_{10} = 49$ (D) $a_n = 4n + 15$, $a_{10} = 55$

ANSWER: (A) $a_n = 4n + 11$, $a_{10} = 51$

CHAPTER 6 - COORDINATE GEOMETRY

=====

CHAPTER: 6 - Coordinate Geometry

=====

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of the point $A(4a, 3a)$ from x-axis is:

OPTIONS: (A) $3a$ (B) $-3a$ (C) $4a$ (D) $-4a$

ANSWER: (A) 3a

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The mid-point of the line segment joining the points (5, -4) and (6, 4) lies on:

OPTIONS: (A) x-axis (B) y-axis (C) origin (D) neither x-axis nor y-axis

ANSWER: (A) x-axis

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the distance between the points (4, p) and (1, 0) is 5, then p is equal to:

OPTIONS: (A) ± 4 (B) 4 (C) -4 (D) 0

ANSWER: (A) ± 4

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A circle centred at (-1, 2) passes through the point (0, 3). Radius of the circle is:

OPTIONS: (A) $2\sqrt{2}$ (B) $\sqrt{2}$ (C) $\sqrt{26}$ (D) 1

ANSWER: (B) $\sqrt{2}$

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The line segment joining the points P(-4, -2) and Q(10, 4) is divided by y-axis in the ratio:

OPTIONS: (A) 2:5 (B) 1:2 (C) 2:1 (D) 5:2

ANSWER: (A) 2:5

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If the points A(4, 5), B(m, 6), C(4, 3) and D(1, n) taken in this order are the vertices of a parallelogram ABCD, then find the values of m and n.

OPTIONS: (A) $m = 7, n = 2$ (B) $m = 8, n = 3$ (C) $m = 6, n = 1$ (D) $m = 7, n = 3$

ANSWER: (A) $m = 7, n = 2$

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the coordinates of the points of trisection of the line segment joining the points A(-1, 4) and B(-3, -2).

OPTIONS: (A) $(-5/3, 2)$ and $(-7/3, 0)$ (B) $(-4/3, 1)$ and $(-5/3, 0)$ (C) $(-2/3, 3)$ and $(-4/3, 1)$ (D) $(-1/3, 4)$ and $(-2/3, 2)$

ANSWER: (A) $(-5/3, 2)$ and $(-7/3, 0)$

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A point P(x, 7) divides a line segment joining the points A(-5, 4) and B(7, 9) in a certain ratio. Find the ratio and hence find the value of x.

OPTIONS: (A) 3:2, $x = 11/5$ (B) 2:3, $x = 11/5$ (C) 3:2, $x = 13/5$ (D) 2:3, $x = 13/5$

ANSWER: (A) 3:2, $x = 11/5$

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The three vertices of a rhombus PQRS are P(2, -3), Q(6, 5) and R(-2, 1). Find the coordinates of the fourth vertex S and coordinates of the point where both the diagonals PR and QS intersect.

OPTIONS: (A) S(-6, -7), mid-point = (0, -1) (B) S(-4, -5), mid-point = (0, -1) (C) S(-6, -7), mid-point = (1, -2) (D) S(-8, -9), mid-point = (0, -1)

ANSWER: (A) S(-6, -7), mid-point = (0, -1)

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: The coordinates of the centre of a circle are $(x - 7, 2x)$. Find the value(s) of 'x', if the circle passes through the point $(-9, 11)$ and has radius $5\sqrt{2}$ units.

OPTIONS: (A) 3, 5 (B) 2, 6 (C) 4, 7 (D) 1, 8

ANSWER: (A) 3, 5

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A point P divides the line segment joining the points A(-3, 5) and B(7, -4) in a certain ratio. If the point P lies on the line $y = 2x$, then find the ratio AP : PB and coordinates of point P.

OPTIONS: (A) 11:18, $(\frac{23}{29}, \frac{46}{29})$ (B) 18:11, $(\frac{23}{29}, \frac{46}{29})$ (C) 11:18, $(\frac{24}{29}, \frac{48}{29})$ (D) 18:11, $(\frac{24}{29}, \frac{48}{29})$

ANSWER: (A) 11:18, $(\frac{23}{29}, \frac{46}{29})$

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Determine the ratio in which the line $3x + y - 9 = 0$ divides the line segment joining the points (1, 3) and (2, 5). Find the point of intersection.

OPTIONS: (A) 3:2, $(\frac{8}{5}, \frac{21}{5})$ (B) 2:3, $(\frac{8}{5}, \frac{21}{5})$ (C) 3:2, $(\frac{7}{5}, \frac{19}{5})$ (D) 2:3, $(\frac{7}{5}, \frac{19}{5})$

ANSWER: (A) 3:2, $(\frac{8}{5}, \frac{21}{5})$

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Do the points P(1, 0), Q(-5, 0) and R(-2, 5) form a triangle? If so, name the type of triangle formed.

OPTIONS: (A) Yes, isosceles triangle (B) Yes, equilateral triangle (C) Yes, scalene triangle (D) No, they are collinear

ANSWER: (A) Yes, isosceles triangle

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Vertices of a right triangle ABC with angle B = 90° are A(3, 4), B(1, 1) and C(-8, 7). Find the value of $\tan A$.

OPTIONS: (A) 3 (B) $\frac{1}{3}$ (C) 2 (D) $\frac{1}{2}$

ANSWER: (A) 3

YEAR: 2025-26

CHAPTER: 6 - Coordinate Geometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Using distance formula, prove that the points A(2, 3), B(-7, 0) and C(-1, 2) are collinear.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

CHAPTER 7 - TRIANGLES

=====

CHAPTER: 7 - Triangles

=====

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In triangles ABC and PQR, angle A = angle Q and angle B = angle R, then AB : AC is equal to:

OPTIONS: (A) PQ : PR (B) PQ : QR (C) QR : QP (D) PR : QR

ANSWER: (C) QR : QP

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: It is given that $\triangle ABC \sim \triangle QRP$ such that $AB = 9$ cm, $BC = 5$ cm and $PR = 2$ cm.

Length of side QR is:

OPTIONS: (A) 0.9 cm (B) $5/18$ cm (C) $10/9$ cm (D) 3.6 cm

ANSWER: (D) 3.6 cm

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Shown below are three triangles. The measures of two adjacent sides and included angle are given for each triangle. Which of these triangles are similar?

OPTIONS: (A) $\triangle RPQ$ and $\triangle XZY$ (B) $\triangle RPQ$ and $\triangle MNL$ (C) $\triangle XZY$ and $\triangle MNL$ (D) $\triangle RPQ$, $\triangle XZY$ and $\triangle MNL$ are similar to one another

ANSWER: (A) $\triangle RPQ$ and $\triangle XZY$

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: It is given that $\triangle ABC \sim \triangle EDF$. Which of the following is not true?

OPTIONS: (A) Perimeter of $\triangle ABC$ / Perimeter of $\triangle EDF = AB/ED$ (B) $AB/ED = AC/EF$ (C) $\angle A = \angle D$, $\angle C = \angle F$ (D) $(AB + BC)/AC = (DE + DF)/EF$

ANSWER: (C) $\angle A = \angle D$, $\angle C = \angle F$

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, $\triangle AHK \sim \triangle ABC$. If $AK = 10$ cm, $BC = 3.5$ cm and $HK = 7$ cm, find the length of AC .

OPTIONS: (A) 5 cm (B) 4 cm (C) 6 cm (D) 7 cm

ANSWER: (A) 5 cm

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, $XY \parallel QR$, $PQ/XQ = 7/3$ and $PR = 6.3$ cm. Find the length of YR .

OPTIONS: (A) 2.7 cm (B) 3.0 cm (C) 2.5 cm (D) 3.3 cm

ANSWER: (A) 2.7 cm

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In $\triangle ABC$, $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, then find the value of x .

OPTIONS: (A) 4 (B) 5 (C) 3 (D) 6

ANSWER: (A) 4

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: As shown in the given figure, a girl of height 90 cm is walking away from the base of a lamp post at a speed of 1.2 m/s. If the lamp is 3.6 m above the ground, find the length of her shadow after 4 seconds.

OPTIONS: (A) 1.6 m (B) 2.0 m (C) 1.8 m (D) 2.2 m

ANSWER: (A) 1.6 m

YEAR: 2025-26

CHAPTER: 7 - Triangles

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In $\triangle ABC$, AD is a median. X is a point on AD such that $AX : XD = 2 : 3$. BX is extended so that it intersects AC at Y. Prove that $BX = 4XY$.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

CHAPTER 8 - CIRCLES

=====

CHAPTER: 8 - Circles

=====

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, PA and PB are tangents to a circle centred at O. If $\angle OAB = 15^\circ$, then $\angle APB$ equals:

OPTIONS: (A) 30° (B) 15° (C) 45° (D) 10°

ANSWER: (A) 30°

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, PA and PB are tangents to a circle centred at O. If $\angle AOB = 130^\circ$, then $\angle APB$ is equal to:

OPTIONS: (A) 130° (B) 50° (C) 120° (D) 90°

ANSWER: (B) 50°

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, PQ and PR are tangents to a circle with centre O and radius 3 cm. If $\angle QPR = 60^\circ$, then the length of each tangent is:

OPTIONS: (A) $3\sqrt{3}$ cm (B) 3 cm (C) 6 cm (D) $\sqrt{3}$ cm

ANSWER: (A) $3\sqrt{3}$ cm

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If PQ and PR are tangents to the circle with centre O and radius 4 cm such that $\angle QPR = 90^\circ$, then the length OP is:

OPTIONS: (A) 4 cm (B) $4\sqrt{2}$ cm (C) 8 cm (D) $2\sqrt{2}$ cm

ANSWER: (B) $4\sqrt{2}$ cm

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, PT is a tangent to the circle with centre O and radius r. If $\angle POT = 45^\circ$, then the length of OP is:

OPTIONS: (A) $r\sqrt{2}$ (B) $\sqrt{2}r$ (C) $2r$ (D) r^2

ANSWER: (A) $r\sqrt{2}$

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: PA and PB are tangents to a circle centred at O. If $\angle PBA = 65^\circ$, then $\angle APB$ equals:

OPTIONS: (A) 65° (B) 60° (C) 50° (D) 35°

ANSWER: (C) 50°

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: PQ is tangent to a circle with centre O. If $\angle POR = 65^\circ$, then $m\angle PTR$ is:

OPTIONS: (A) 65° (B) 58.5° (C) 57.5° (D) 45°

ANSWER: (C) 57.5°

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that the lengths of tangents drawn from an external point to a circle are equal.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-26

CHAPTER: 8 - Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If TP and TQ are two tangents to a circle with centre O from an external point T so that $\angle POQ = 120^\circ$, then $\angle PTQ$ is equal to:

OPTIONS: (A) 60° (B) 70° (C) 80° (D) 90°

ANSWER: (A) 60°

CHAPTER 9 - INTRODUCTION TO TRIGONOMETRY

=====

CHAPTER: 9 - Introduction to Trigonometry

=====

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\cos A = 4/5$, then the value of $\tan A$ is:

OPTIONS: (A) $3/5$ (B) $3/4$ (C) $4/3$ (D) $5/3$

ANSWER: (B) $3/4$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $2 \sin A = 1$, then the value of $\tan A + \cot A$ is:

OPTIONS: (A) $\sqrt{3}$ (B) $4/\sqrt{3}$ (C) $\sqrt{3}/2$ (D) 1

ANSWER: (B) $4/\sqrt{3}$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Given $\cot \theta = 3$, the value of $\cos \theta$ is:

OPTIONS: (A) $1/3$ (B) $1/\sqrt{10}$ (C) $3/\sqrt{10}$ (D) $\sqrt{10}/3$

ANSWER: (C) $3/\sqrt{10}$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Given that $\sin \theta = a/b$, then $\cos \theta$ is equal to:

OPTIONS: (A) $b/\sqrt{b^2 - a^2}$ (B) b/a (C) $\sqrt{b^2 - a^2}/b$ (D) $a/\sqrt{b^2 - a^2}$

ANSWER: (C) $\sqrt{b^2 - a^2}/b$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\cos A = 1/2$, then the value of $\sin^2 A + 2 \cos^2 A$ is:

OPTIONS: (A) $3/2$ (B) $5/4$ (C) -1 (D) $1/2$

ANSWER: (B) $5/4$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Given that $\sin 2\alpha = \sqrt{3}/2$, the value of $\sin 3\alpha$ is:

OPTIONS: (A) $3\sqrt{3}/4$ (B) $1/2$ (C) 1 (D) $\sqrt{3}/4$

ANSWER: (C) 1

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(1/2 \tan^2 45^\circ - \cos^2 60^\circ)$ is:

OPTIONS: (A) 0 (B) $-1/2$ (C) $1/4$ (D) $-1/4$

ANSWER: (C) $1/4$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: $(1 + \tan^2 A)/(1 + \cot^2 A)$ equals to:

OPTIONS: (A) $\tan^2 A$ (B) -1 (C) $-\tan^2 A$ (D) $\cot^2 A$

ANSWER: (A) $\tan^2 A$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $2 \tan A = 3$, then value of $\sec A$ equals:

OPTIONS: (A) $\sqrt{13}/2$ (B) $\sqrt{13}/4$ (C) $2/\sqrt{13}$ (D) $\sqrt{13}/2$

ANSWER: (D) $\sqrt{13}/2$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\cos y = 0$, then what is the value of $1/2 \cos y/2$?

OPTIONS: (A) 0 (B) $1/2$ (C) $1/\sqrt{2}$ (D) $1/(2\sqrt{2})$

ANSWER: (D) $1/(2\sqrt{2})$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When $\sin A = 1/3$, the value of $\cot A$ is:

OPTIONS: (A) $2\sqrt{2}/3$ (B) $2\sqrt{2}$ (C) $1/(2\sqrt{2})$ (D) 3

ANSWER: (B) $2\sqrt{2}$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $\tan \theta = 24/7$, then find the value of $\sin \theta + \cos \theta$.

OPTIONS: (A) $31/25$ (B) $25/31$ (C) $31/24$ (D) $24/25$

ANSWER: (A) $31/25$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $\cot \theta = 7/8$, then find the value of $(1 + \sin \theta)(1 - \sin \theta)/(1 + \cos \theta)(1 - \cos \theta)$.

OPTIONS: (A) $49/64$ (B) $64/49$ (C) $7/8$ (D) $8/7$

ANSWER: (A) $49/64$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(\sin A + \sec A)^2 + (\cos A + \operatorname{cosec} A)^2 = (1 + \sec A \operatorname{cosec} A)^2$

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \sec \theta - \tan \theta$

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If $\sin \theta + \cos \theta = \sqrt{3}$, then prove that $\tan \theta + \cot \theta = 1$.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\tan \theta / (1 - \cot \theta) + \cot \theta / (1 - \tan \theta) = 1 + \tan \theta + \cot \theta$.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If $\tan A = 4/3$, find $\sin A$ and $\cos A$.

OPTIONS: (A) $\sin A = 4/5$, $\cos A = 3/5$ (B) $\sin A = 3/5$, $\cos A = 4/5$ (C) $\sin A = 4/5$, $\cos A = 4/5$
(D) $\sin A = 3/5$, $\cos A = 3/5$

ANSWER: (A) $\sin A = 4/5$, $\cos A = 3/5$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Express $\cos A$ and $\tan A$ in terms of $\sin A$.

OPTIONS: (A) $\cos A = \sqrt{1 - \sin^2 A}$, $\tan A = \sin A / \sqrt{1 - \sin^2 A}$ (B) $\cos A = \sqrt{1 + \sin^2 A}$, $\tan A = \sin A / \sqrt{1 + \sin^2 A}$ (C) $\cos A = \sqrt{1 - \sin^2 A}$, $\tan A = \sqrt{1 - \sin^2 A} / \sin A$ (D) $\cos A = \sqrt{1 + \sin^2 A}$, $\tan A = \sqrt{1 + \sin^2 A} / \sin A$

ANSWER: (A) $\cos A = \sqrt{1 - \sin^2 A}$, $\tan A = \sin A / \sqrt{1 - \sin^2 A}$

YEAR: 2025-26

CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If $\sec \theta + \tan \theta = m$, show that $(m^2 - 1)/(m^2 + 1) = \sin \theta$.

OPTIONS: (A) True (B) False (C) Cannot be determined (D) Partially true

ANSWER: (A) True

CHAPTER 10 - AREAS RELATED TO CIRCLES

=====

CHAPTER: 10 - Areas Related to Circles

=====

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Area of a segment of a circle of radius 'r' and central angle 60° is:

OPTIONS: (A) $\pi r^2/2 - 1/2 r^2$ (B) $2\pi r/4 - \sqrt{3}/4 r^2$ (C) $\pi r^2/6 - \sqrt{3}/4 r^2$ (D) $2\pi r/4 - r^2 \sin 60^\circ$

ANSWER: (C) $\pi r^2/6 - \sqrt{3}/4 r^2$

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The length of the arc of the sector of a circle with radius 21 cm and of central angle 60° , is:

OPTIONS: (A) 22 cm (B) 44 cm (C) 88 cm (D) 11 cm

ANSWER: (A) 22 cm

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A circle is divided into 16 identical sectors. If radius of the circle is 7 cm, area of each sector is:

OPTIONS: (A) $77/4 \text{ cm}^2$ (B) 77 cm^2 (C) 154 cm^2 (D) $77/8 \text{ cm}^2$

ANSWER: (D) $77/8 \text{ cm}^2$

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Area of sector of a circle with radius 18 cm is 198 cm^2 . The measure of central angle is:

OPTIONS: (A) 70° (B) 14° (C) 140° (D) 210°

ANSWER: (A) 70°

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The area of a sector of a circle of radius 10 cm is $55/3 \text{ cm}^2$. The value of central angle is:

OPTIONS: (A) $21^\circ/2$ (B) 42° (C) 105° (D) 21°

ANSWER: (D) 21°

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the area of the sector of a circle of radius 42 cm and of central angle 30° .

Also, find the area of the corresponding major sector. [Use $\pi = \frac{22}{7}$]

OPTIONS: (A) Minor sector = 462 cm^2 , Major sector = 5082 cm^2 (B) Minor sector = 462 cm^2 , Major sector = 4620 cm^2 (C) Minor sector = 462 cm^2 , Major sector = 5082 cm^2 (D) Minor sector = 462 cm^2 , Major sector = 5082 cm^2

ANSWER: (A) Minor sector = 462 cm^2 , Major sector = 5082 cm^2

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A circle of diameter 20 cm is equally divided into five sectors. Find the area and perimeter of one of the sectors.

OPTIONS: (A) Area = $\frac{440}{7} \text{ cm}^2$, Perimeter = $\frac{228}{7} \text{ cm}$ (B) Area = $\frac{440}{7} \text{ cm}^2$, Perimeter = $\frac{228}{7} \text{ cm}$ (C) Area = $\frac{440}{7} \text{ cm}^2$, Perimeter = $\frac{228}{7} \text{ cm}$ (D) Area = $\frac{440}{7} \text{ cm}^2$, Perimeter = $\frac{228}{7} \text{ cm}$

ANSWER: (A) Area = $\frac{440}{7} \text{ cm}^2$, Perimeter = $\frac{228}{7} \text{ cm}$

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the area of the segment AYB shown in the figure, if the radius of the circle is 21 cm and $\angle AOB = 120^\circ$. [Use $\pi = \frac{22}{7}$]

OPTIONS: (A) $462 - 441\sqrt{\frac{3}{4}} \text{ cm}^2$ (B) $462 - 441\sqrt{\frac{3}{2}} \text{ cm}^2$ (C) $462 - 441\sqrt{\frac{3}{4}} \text{ cm}^2$ (D) $462 - 441\sqrt{\frac{3}{2}} \text{ cm}^2$

ANSWER: (A) $462 - 441\sqrt{\frac{3}{4}} \text{ cm}^2$

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, chord AB subtends an angle of 120° at the centre of the circle with radius 7 cm. Find (i) perimeter of major sector OACB, and (ii) area of the shaded segment, if area of $\triangle OAB = 21.2 \text{ cm}^2$.

OPTIONS: (A) Perimeter = $130/3 \text{ cm}$, Area = 30.1 cm^2 (B) Perimeter = $130/3 \text{ cm}$, Area = 30.1 cm^2 (C) Perimeter = $130/3 \text{ cm}$, Area = 30.1 cm^2 (D) Perimeter = $130/3 \text{ cm}$, Area = 30.1 cm^2

ANSWER: (A) Perimeter = $130/3 \text{ cm}$, Area = 30.1 cm^2

YEAR: 2025-26

CHAPTER: 10 - Areas Related to Circles

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A chord of a circle, of radius 14 cm, subtends an angle of 60° at the centre. Find the area of the smaller sector and perimeter of the smaller segment.

OPTIONS: (A) Area = $308/3 \text{ cm}^2$, Perimeter = $86/3 \text{ cm}$ (B) Area = $308/3 \text{ cm}^2$, Perimeter = $86/3 \text{ cm}$ (C) Area = $308/3 \text{ cm}^2$, Perimeter = $86/3 \text{ cm}$ (D) Area = $308/3 \text{ cm}^2$, Perimeter = $86/3 \text{ cm}$

ANSWER: (A) Area = $308/3 \text{ cm}^2$, Perimeter = $86/3 \text{ cm}$

CHAPTER 11 - SURFACE AREAS AND VOLUMES

=====

CHAPTER: 11 - Surface Areas and Volumes

=====

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A hemispherical bowl is made of steel of thickness 1 cm. The outer radius of the bowl is 6 cm. The volume of steel used (in cm^3) is:

OPTIONS: (A) 182π (B) $182/3 \pi$ (C) $682/3 \pi$ (D) $364/3 \pi$

ANSWER: (B) $182/3 \pi$

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The total surface area of a solid hemisphere of diameter '2d' is:

OPTIONS: (A) $3\pi d^2$ (B) $2\pi d^2$ (C) $\frac{1}{2}\pi d^2$ (D) $\frac{3}{4}\pi d^2$

ANSWER: (A) $3\pi d^2$

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A cone of maximum size is carved out from a solid cube of edge length l. The volume of the cone is:

OPTIONS: (A) $\frac{\pi l^3}{12}$ (B) $\frac{\pi l^3}{3}$ (C) $l^3(1 - \frac{\pi}{3})$ (D) $\frac{\pi l^3}{8}$

ANSWER: (A) $\frac{\pi l^3}{12}$

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Three tennis balls are just packed in a cylindrical jar. If radius of each ball is r, volume of air inside the jar is:

OPTIONS: (A) $2\pi r^3$ (B) $3\pi r^3$ (C) $5\pi r^3$ (D) $4\pi r^3$

ANSWER: (A) $2\pi r^3$

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: An ice-cream cone of radius r and height h is completely filled by two spherical scoops of ice-cream. If radius of each spherical scoop is $\frac{r}{2}$, then h : 2r equals:

OPTIONS: (A) 1:8 (B) 1:2 (C) 1:1 (D) 2:1

ANSWER: (B) 1:2

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A toy is in the form of a cone mounted on a hemisphere of radius 7 cm. The total height of the toy is 31 cm. Find the total surface area of the toy.

OPTIONS: (A) 858 cm^2 (B) 858 cm^2 (C) 858 cm^2 (D) 858 cm^2

ANSWER: (A) 858 cm^2

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: 1000 small thermocold balls of radius 0.5 cm are kept in a spherical balloon of radius 20 cm. Find the volume of air in the balloon.

OPTIONS: (A) 33000 cm^3 (B) 33000 cm^3 (C) 33000 cm^3 (D) 33000 cm^3

ANSWER: (A) 33000 cm^3

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A solid is in the form of a cylinder with hemispherical ends. The total height of the solid is 20 cm and the diameter of the cylinder is 7 cm. Find the total volume of the solid. (Use $\pi = \frac{22}{7}$)

OPTIONS: (A) 902.33 cm^3 (B) 902.33 cm^3 (C) 902.33 cm^3 (D) 902.33 cm^3

ANSWER: (A) 902.33 cm^3

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A conical cavity of maximum volume is carved out from a wooden solid hemisphere of radius 10 cm. Curved surface area of the cavity carved out is (use $\pi = 3.14$):

OPTIONS: (A) $314\sqrt{2} \text{ cm}^2$ (B) 314 cm^2 (C) $3140/3 \text{ cm}^2$ (D) $3140\sqrt{2} \text{ cm}^2$

ANSWER: (A) $314\sqrt{2} \text{ cm}^2$

YEAR: 2025-26

CHAPTER: 11 - Surface Areas and Volumes

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A camping tent in hemispherical shape of radius 1.4 m, has a door opening of area 0.50 m^2 . Outer surface area of the tent is:

OPTIONS: (A) 11.78 m^2 (B) 12.32 m^2 (C) 11.82 m^2 (D) 12.86 m^2

ANSWER: (C) 11.82 m^2

CHAPTER 12 - STATISTICS

=====

CHAPTER: 12 - Statistics

=====

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The mean and median of a frequency distribution are 43 and 43.4 respectively. The mode of the distribution is:

OPTIONS: (A) 43.4 (B) 42.4 (C) 44.2 (D) 49.3

ANSWER: (C) 44.2

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The median and mode of a distribution are 25.2 and 26.1 respectively. The mean of the distribution is:

OPTIONS: (A) 24.75 (B) 24.25 (C) 24.3 (D) 25.5

ANSWER: (A) 24.75

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Mean and Median of a frequency distribution are 43 and 40 respectively. The value of mode is:

OPTIONS: (A) 34 (B) 43 (C) 38.5 (D) 41.5

ANSWER: (A) 34

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: An SBI health insurance agent found the following data for distribution of ages of 100 policy holders. The health insurance policies are given to persons of age 15 years and onwards, but less than 60 years. Find the modal age and median age of the policy holders.

OPTIONS: (A) Mode = 36.7 years, Median = 35.7 years (B) Mode = 37.2 years, Median = 36.1 years (C) Mode = 35.8 years, Median = 34.9 years (D) Mode = 36.7 years, Median = 35.7 years

ANSWER: (A) Mode = 36.7 years, Median = 35.7 years

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The median of the following data is 137. Find the values of x and y, given that total of frequencies is 68.

OPTIONS: (A) $x = 13$, $y = 8$ (B) $x = 12$, $y = 9$ (C) $x = 14$, $y = 7$ (D) $x = 15$, $y = 6$

ANSWER: (A) $x = 13$, $y = 8$

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: If the median of the following distribution is 32.5, then find the values of x and y.

OPTIONS: (A) $x = 3$, $y = 6$ (B) $x = 4$, $y = 5$ (C) $x = 2$, $y = 7$ (D) $x = 5$, $y = 4$

ANSWER: (A) $x = 3$, $y = 6$

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The mean of the following frequency distribution is 28. If sum of all frequencies is 100, then find the values of p and q.

OPTIONS: (A) $p = 18$, $q = 17$ (B) $p = 17$, $q = 18$ (C) $p = 20$, $q = 15$ (D) $p = 15$, $q = 20$

ANSWER: (A) $p = 18$, $q = 17$

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The mean of the following frequency distribution is 35. Find the values of x and y , if the sum of frequencies is 25.

OPTIONS: (A) $x = 8$, $y = 7$ (B) $x = 7$, $y = 8$ (C) $x = 9$, $y = 6$ (D) $x = 6$, $y = 9$

ANSWER: (A) $x = 8$, $y = 7$

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find the mean and mode of the following distribution.

OPTIONS: (A) Mean = 34.8, Mode = 42.3 (B) Mean = 35.2, Mode = 41.8 (C) Mean = 34.8, Mode = 42.3 (D) Mean = 35.2, Mode = 41.8

ANSWER: (A) Mean = 34.8, Mode = 42.3

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find mean and mode of the following frequency distribution.

OPTIONS: (A) Mean = 33.4, Mode = 31.25 (B) Mean = 34.2, Mode = 32.5 (C) Mean = 33.4, Mode = 31.25 (D) Mean = 34.2, Mode = 32.5

ANSWER: (A) Mean = 33.4, Mode = 31.25

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: If the median of the distribution given below is 28.5, find the values of x and y .

OPTIONS: (A) $x = 8$, $y = 7$ (B) $x = 7$, $y = 8$ (C) $x = 9$, $y = 6$ (D) $x = 6$, $y = 9$

ANSWER: (A) $x = 8$, $y = 7$

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find mean and mode of the following frequency distribution.

OPTIONS: (A) Mean = 33.4, Mode = 31.25 (B) Mean = 33.4, Mode = 31.25 (C) Mean = 33.4, Mode = 31.25 (D) Mean = 33.4, Mode = 31.25

ANSWER: (A) Mean = 33.4, Mode = 31.25

YEAR: 2025-26

CHAPTER: 12 - Statistics

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: While calculating mean of a grouped frequency distribution, step deviation method was used $(x - a)/h = u$. It was found that $\bar{x} = 64$, $h = 5$ and $a = 62.5$. The value of $\bar{u}$ is:

OPTIONS: (A) 0.5 (B) 1.5 (C) 0.3 (D) 7.5

ANSWER: (C) 0.3

CHAPTER 13 - PROBABILITY

=====

CHAPTER: 13 - Probability

=====

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The probability for a randomly selected number out of 1, 2, 3, 4, ..., 25 to be a composite number is:

OPTIONS: (A) 15/25 (B) 10/25 (C) 11/25 (D) 9/25

ANSWER: (A) 15/25

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two dice are rolled together. The probability that the sum of the numbers obtained is divisible by 6, is:

OPTIONS: (A) $\frac{1}{6}$ (B) $\frac{11}{36}$ (C) $\frac{1}{12}$ (D) $\frac{1}{4}$

ANSWER: (A) $\frac{1}{6}$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A die is thrown once. Probability of getting a number other than 3 is:

OPTIONS: (A) $\frac{1}{6}$ (B) $\frac{3}{6}$ (C) $\frac{5}{6}$ (D) 1

ANSWER: (C) $\frac{5}{6}$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two dice are rolled together. The probability that both the obtained numbers are less than 4, is:

OPTIONS: (A) $\frac{2}{9}$ (B) $\frac{7}{36}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$

ANSWER: (C) $\frac{1}{4}$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A card is drawn at random from a well shuffled deck of 52 playing cards. The probability that it is either a ten or a king is:

OPTIONS: (A) $\frac{1}{26}$ (B) $\frac{2}{13}$ (C) $\frac{1}{13}$ (D) $\frac{8}{26}$

ANSWER: (B) $\frac{2}{13}$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following can not be the probability of an event?

OPTIONS: (A) $39/100$ (B) $0.001/20$ (C) $10/0.2$ (D) 10%

ANSWER: (C) $10/0.2$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two dice are rolled together. The probability that sum of the numbers obtained is atmost 12, is:

OPTIONS: (A) 1 (B) 0 (C) $1/2$ (D) $35/36$

ANSWER: (A) 1

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Two dice are thrown at the same time. Determine the probability that the (i) sum of the numbers on the two dice is 5, and (ii) difference of the numbers on the two dice is 3.

OPTIONS: (A) (i) $4/36$, (ii) $6/36$ (B) (i) $5/36$, (ii) $7/36$ (C) (i) $4/36$, (ii) $6/36$ (D) (i) $5/36$, (ii) $7/36$

ANSWER: (A) (i) $4/36$, (ii) $6/36$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Two different dice are thrown together. Find the probability that the numbers obtained have: (i) even sum, (ii) even product.

OPTIONS: (A) (i) $1/2$, (ii) $3/4$ (B) (i) $1/2$, (ii) $3/4$ (C) (i) $1/2$, (ii) $3/4$ (D) (i) $1/2$, (ii) $3/4$

ANSWER: (A) (i) $1/2$, (ii) $3/4$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Two different coins are tossed simultaneously. What is the probability of getting: (i) at least one head? (ii) at most one tail? (iii) a head and a tail?

OPTIONS: (A) (i) $\frac{3}{4}$, (ii) $\frac{3}{4}$, (iii) $\frac{1}{2}$ (B) (i) $\frac{3}{4}$, (ii) $\frac{3}{4}$, (iii) $\frac{1}{2}$ (C) (i) $\frac{3}{4}$, (ii) $\frac{3}{4}$, (iii) $\frac{1}{2}$ (D) (i) $\frac{3}{4}$, (ii) $\frac{3}{4}$, (iii) $\frac{1}{2}$

ANSWER: (A) (i) $\frac{3}{4}$, (ii) $\frac{3}{4}$, (iii) $\frac{1}{2}$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A bag contains 30 balls out of which 'm' number of balls are blue in colour. (i) Find the probability that a ball drawn at random from the bag is not blue. (ii) If 6 more blue balls are added in the bag, then the probability of drawing a blue ball will be $\frac{5}{4}$ times the probability of drawing a blue ball in the first case. Find the value of m.

OPTIONS: (A) (i) $\frac{(30-m)}{30}$, (ii) $m = 12$ (B) (i) $\frac{(30-m)}{30}$, (ii) $m = 12$ (C) (i) $\frac{(30-m)}{30}$, (ii) $m = 12$ (D) (i) $\frac{(30-m)}{30}$, (ii) $m = 12$

ANSWER: (A) (i) $\frac{(30-m)}{30}$, (ii) $m = 12$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Meena calculates that the probability of her winning the first prize in a lottery is 0.08. If total 800 tickets were sold, the number of tickets bought by her, is:

OPTIONS: (A) 64 (B) 640 (C) 100 (D) 10

ANSWER: (A) 64

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A bag contains 25 balls. Some of them are yellow and others are green. One ball is drawn at random. If probability of getting a green ball is $\frac{3}{5}$, then find the number of yellow balls.

OPTIONS: (A) 10 (B) 12 (C) 15 (D) 8

ANSWER: (A) 10

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two dice are rolled together. The probability of getting an outcome (x, y) where $x > y$, is:

OPTIONS: (A) $5/12$ (B) $5/6$ (C) 1 (D) 0

ANSWER: (A) $5/12$

YEAR: 2025-26

CHAPTER: 13 - Probability

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A bag contains 30 balls out of which 'm' number of balls are blue in colour. (i) Find the probability that a ball drawn at random from the bag is not blue. (ii) If 6 more blue balls are added in the bag, then the probability of drawing a blue ball will be $5/4$ times the probability of drawing a blue ball in the first case. Find the value of m.

OPTIONS: (A) (i) $(30-m)/30$, (ii) $m = 12$ (B) (i) $(30-m)/30$, (ii) $m = 12$ (C) (i) $(30-m)/30$, (ii) $m = 12$ (D) (i) $(30-m)/30$, (ii) $m = 12$

ANSWER: (A) (i) $(30-m)/30$, (ii) $m = 12$